Exercises

Exercise: Linear model with single categorical explanatory variable

1. As in previous exercises, either create a new R script (perhaps call it 'linear_model_2') or continue with your previous R script in your RStudio Project. Again, make sure you include any metadata you feel is appropriate (title, description of task, date of creation etc) and don't forget to comment out your metadata with a `#` at the beginning of the line.

2. Once again import the data file 'loyn.txt' into R and take a look at the structure of this dataframe using the `str()` function. In this exercise you will investigate whether the abundance of birds (`ABUND`) is different in areas with different grazing intensities (`GRAZE`). Remember, the `GRAZE` variable is an index of livestock grazing intensity. Level 1 = low grazing intensity and level 5 = high grazing intensity.

3. As we discussed in the graphical data exploration exercise the `GRAZE` variable was originally coded as a numeric (i.e. 1, 2, 3, 4, 5). In this exercise we actually want to treat `GRAZE` as a categorical variable with five levels (aka a factor). So the first thing we need to do is create a new variable in the `loyn` dataframe called `FGRAZE` in which we store the `GRAZE` variable coerced to be a categorical variable with the `factor()` function (you can also use the `as.factor()` function if you prefer).

4. Explore any potential differences in bird abundance between each level of `FGRAZE` graphically using an appropriate plot (hint: a boxplot might be useful here). How would you interpret this plot? What might you expect to see in your analysis? Write your predictions in your R script as a comment. What is the mean number of birds for each level of `FGRAZE`?

5. Fit an appropriate linear model in R to explain the variation in the response variable, `ABUND`, with the explanatory variable `FGRAZE`. Remember to use the `data =` argument. Assign this linear model to an appropriately named object (`birds_lm` if your imagination fails you!).

6. Produce the ANOVA table using the `anova()` function on the model object. What null hypothesis is being tested? Do you reject or fail to reject the null hypothesis? What summary statistics would you report? Summarise in your R script as a comment.

7. Use the `summary()` function on the model object to produce the table of parameter estimates (remember these are called coefficients in R). Using this output what is the estimate of the intercept and what does this represent? What is the null hypothesis associated with the intercept? do you reject or fail to reject this hypothesis?

Next we move onto the FGRAZE2 parameter, how do you interpret this parameter? (remember they are contrasts). Again, what is the null hypothesis associated with the FGRAZE2 parameter? do you reject or fail to reject this hypothesis?

Repeat this interpretation for the FGRAZE3, FGRAZE4 and FGRAZE5 parameters. Summarise this as a comment in your R script.

Finally, just a reminder about P values and confidence intervals (see the note in the previous exercise if you'd like it again): lead with the estimate and its interval rather than with the test, and never read a large P value as evidence that two things are the same. Use the `confint()` function on your model object to obtain the 95% confidence intervals for the intercept and for each of the four contrasts. Now go back over what you've just written and report each contrast again as an estimated difference in bird abundance *with* its confidence interval, rather than as a hypothesis that you either reject or fail to reject. Which of the four *differences* has been estimated most precisely, and which least precisely? Can you see why? (hint: each contrast compares two grazing levels, so how precisely you can estimate it depends on how many forest patches there are in **both** of them. The `table()` function will give you the group sizes).

8. Now that you have interpreted all the contrasts with FGRAZE level 1 as the intercept, set the intercept to FGRAZE level 2 using the `relevel()` function, refit the model, produce the new table of parameter estimates using the `summary()` function again and interpret.

Repeat this for FGRAZE levels 3, 4 and 5. Can you summarise which levels of FGRAZE are different from each other? Use `confint()` on each of your refitted models as well as `summary()`, and as you go, keep an eye on what releveling does and does not change.

9. Staying with the summary table of parameter estimates, how much of the variation in bird abundance does the explanatory variable FGRAZE explain? As in the previous exercise, decide which of the two R^2 values you should quote and why. This model estimates more parameters than the last one did, so is the gap between them bigger or smaller here?

10. Now onto a really important part of the model fitting process. Let's check the assumptions of your linear model by creating plots of the residuals from the model. Remember, you can easily create these plots by using the `plot()` function on your model object. Also remember that if you want to see all plots at once then you should split your plotting device into 2 rows and 2 columns using the `par()` function before you create the plots. Check each of the assumptions using these plots and report whether your model meets these assumptions in your R script.

11. To finish, let's plot the model. Don't panic about the code, I'll give you that in the solutions, but do spend some time on the two questions underneath because they matter rather more than the code does.

Have a go at using our old trusty `predict()` function to calculate the fitted mean bird abundance for each level of FGRAZE along with the standard errors. Then plot these fitted values against grazing level and add the 95% confidence intervals (you will need either the `arrows()` or the `segments()` function to add the intervals to the plot). There are also packages that will do the whole job for you in a line or two, and I've put a couple of those in the solutions, so feel free to use Google (yep, this is OK!) and have a look at the `effects` package or `ggplot2` if you fancy it.

Once you have your plot:

a) The five intervals on your plot are **not** the four intervals you calculated in Question 7. What is each one an interval for? And why are there five here but only four in the `summary()` table? Compare the interval for graze level 2 on your plot against the FGRAZE2 interval from `confint()`, and make sure you can say in words why the two are different.

b) Looking at your plot, is it safe to conclude that two grazing levels differ whenever their intervals don't overlap? And is it safe to conclude that they don't differ whenever their intervals do overlap? Think carefully about this one before you look at the solution, because reading differences off overlapping error bars by eye is a mistake that appears in print constantly.

Two other ways of producing the same plot. Both give you essentially the same figure as the one above. These are here purely for information, so that you have some options and some working code if you want to make this kind of plot with your own data.

12. Now for reporting your results. Same idea as the previous exercise, but this time I'm not giving you the sentences, only the ingredients. Write a short results paragraph (four or five sentences is plenty) describing the effect of grazing intensity on bird abundance. Make sure it contains:

- the overall test of whether abundance differs among grazing levels, with its F statistic and degrees of freedom
- the estimated mean abundance at the lowest grazing intensity
- the size of the difference between the lowest and the highest grazing intensities, with its confidence interval
- the sample size

One thing to think about while you're writing. Your `confint()` output hands you four contrasts, all against graze level 1, and writing all four out in prose is tedious to read and buries the pattern. Have another look at your plot from Question 11. What is the pattern here, and can you get it across in a single sentence?

End of the linear model with single categorical explanatory variable exercise